MSG No. 176.144 $P6_3/m1'$ [Type II, hexagonal]

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det	TR
1	$\{1\|0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
2	$\{6^+_{001}\|00\frac{1}{2}\}$	$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	1	1
3	$\{3^+_{001}\|0\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
4	$\{2_{001}\|00\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	1	1
5	$\{3^-_{001}\|0\}$	$\begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
6	$\{6^-_{001}\|00\frac{1}{2}\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	1	1
7	$\{-1\|0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	-1	1
8	$\{-6^+_{001}\|00\frac{1}{2}\}$	$\begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	-1	1
9	$\{-3^+_{001}\|0\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	-1	1
10	$\{\mathrm{m}_{001}\|00\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	-1	1
11	$\{-3^-_{001}\|0\}$	$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	-1	1
12	$\{-6^-_{001}\|00\frac{1}{2}\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	-1	1
13	$\{1'\|0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	-1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
14	$\{6^+_{001}{}'\|00\frac{1}{2}\}$	$\begin{bmatrix}1 & -1 & 0 & 0\\ 1 & 0 & 0 & 0\\ 0 & 0 & 1 & \frac{1}{2}\end{bmatrix}$	1	−1
15	$\{3^+_{001}{}'\|0\}$	$\begin{bmatrix}0 & -1 & 0 & 0\\ 1 & -1 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}$	1	−1
16	$\{2_{001}{}'\|00\frac{1}{2}\}$	$\begin{bmatrix}-1 & 0 & 0 & 0\\ 0 & -1 & 0 & 0\\ 0 & 0 & 1 & \frac{1}{2}\end{bmatrix}$	1	−1
17	$\{3^-_{001}{}'\|0\}$	$\begin{bmatrix}-1 & 1 & 0 & 0\\ -1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}$	1	−1
18	$\{6^-_{001}{}'\|00\frac{1}{2}\}$	$\begin{bmatrix}0 & 1 & 0 & 0\\ -1 & 1 & 0 & 0\\ 0 & 0 & 1 & \frac{1}{2}\end{bmatrix}$	1	−1
19	$\{-1'\|0\}$	$\begin{bmatrix}-1 & 0 & 0 & 0\\ 0 & -1 & 0 & 0\\ 0 & 0 & -1 & 0\end{bmatrix}$	−1	−1
20	$\{-6^+_{001}{}'\|00\frac{1}{2}\}$	$\begin{bmatrix}-1 & 1 & 0 & 0\\ -1 & 0 & 0 & 0\\ 0 & 0 & -1 & \frac{1}{2}\end{bmatrix}$	−1	−1
21	$\{-3^+_{001}{}'\|0\}$	$\begin{bmatrix}0 & 1 & 0 & 0\\ -1 & 1 & 0 & 0\\ 0 & 0 & -1 & 0\end{bmatrix}$	−1	−1
22	$\{\mathrm{m}_{001}{}'\|00\frac{1}{2}\}$	$\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & -1 & \frac{1}{2}\end{bmatrix}$	−1	−1
23	$\{-3^-_{001}{}'\|0\}$	$\begin{bmatrix}1 & -1 & 0 & 0\\ 1 & 0 & 0 & 0\\ 0 & 0 & -1 & 0\end{bmatrix}$	−1	−1
24	$\{-6^-_{001}{}'\|00\frac{1}{2}\}$	$\begin{bmatrix}0 & -1 & 0 & 0\\ 1 & -1 & 0 & 0\\ 0 & 0 & -1 & \frac{1}{2}\end{bmatrix}$	−1	−1